

AquaSDR · FEASIBILITY AND VIABILITY

Proven on our hardware · risks identified · a mitigation for each

ANALYSIS OF FEASIBILITY

Readiness of every subsystem, from design to instrument validation

	DESIGNED	SIMULATED / TESTED	BUILT	ON-BOARD VERIFIED	INSTRUMENT VALIDATED
Adaptive firmware engine C/C++ on STM32F446RE					
Timer + DMA + DAC streaming TIM6 → DMA1 → DAC1 · 1 MS/s					
LFM · geometric · Barker-13 with Hann / Hamming / Blackman					
Sensor input + adaptation CLEAR / TRANSITION / MURKY					
Analog front end Sallen-Key LPF + TLV9062					
Payload enclosure acrylic hull · printed AUV pod				n/a	n/a
Low-power operation DMA-driven, CPU free per sample			n/a		
reached partly (A0 shown in CLEAR; printed pod next) next					

PROVEN ON THE BOARD · MCU ADC capture and firmware telemetry, not oscilloscope data

42.00 kHz captured centre = calculated LFM · Hann	-14.6 → -35 dB spectral leakage, rect → Hann window effect captured	-21.4 dB Barker-13 sidelobe captured -21.7 dB calculated
330 / 330 commands acknowledged 0 lost	0 DMA underruns across every capture	< 100 ms decision → new waveform 86–97 ms measured

VIABILITY	Low-cost COTS STM32 Nucleo · TLV9062 · standard passives	Software-defined new waveform or policy = firmware update	Scalable same architecture → 100–500 kHz with an external DAC	Maintainable 161 automated tests guard every change
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CHALLENGES & RISKS → STRATEGIES

Each risk paired with how we overcome it

CHALLENGE	STRATEGY	STATUS
Frequency scale PS cites 100–500 kHz; bench transducer is 40 kHz	→ Frequency-agnostic policy table; external high-speed DAC on the same TIM + DMA path	SCALE PATH
No oscilloscope on hand external validation not yet possible	→ On-board ADC + DMA capture with FFT and matched filter now; lab scope at judging, CSV import ready	IN PLACE
Analog stage not yet built filter + op-amp still to assemble	→ Two independent models agree within 0.02 dB in band; stock-value parts list ready	NEXT BUILD
Noisy or drifting sensor input a bad reading could change the ping	→ Smoothing, hysteresis, stale and out-of-range rejection; TDS labelled as a proxy	IN PLACE
CPU load and battery budget per-sample trig math is expensive	→ Next pulse computed between pings, DMA streams it; inline-meter power test planned	IN PLACE
Loss of control link ping must never run unattended	→ Output off at boot, 2 s control lease auto-stop, DAC parked mid-scale on any fault	IN PLACE
Water ingress payload must survive the hull slot	→ O-ring end caps; parametric pod compiled and fit-checked in OpenSCAD	DESIGNED

